MIL-M-45105D(MU)
20 December 1966
SUPERSEDING
MIL-M-45105C(MU)
18 September 1963

MILITARY SPECIFICATION

MOUNTS, PERISCOPE: M104, M104A1 AND M104A2

1. SCOPE

1.1 This specification covers three similar types of periscope mounts which employ cylindrical rotary type periscope locks.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F8270307	Mount, Periscope: M104A2
F8270627	Mount, Periscope: M104A1
F8567570	Mount, Periscope: M104

PACKAGING DATA SHEET

8276627	Packaging of Mounts, Periscope M104, M104A1 and M104A2

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.2 Fabrication.- The mount shall be manufactured in accordance with the applicable drawing and drawings pertaining thereto.

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 Operating temperatures.- The mount shall meet the requirement of 3.5 at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) and while exposed to and thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit.

3.5 Functional operation.- Function of the mount shall be evaluated by utilizing a medium representive of a periscope of the M28 series, in size and weight for insertion and retention in the mount. The spring lock of the mount shall seat and hold the medium when the handle of the mount is actuated to the locked position. When the handle of the mount is moved to the unlocked position the spring actuated catch shall prevent the medium from falling free of the mount.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be

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as listed in MIL-STD-109.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examination and tests for Packaging, Packing, and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval prior to Government acceptance or rejection by item or lot.

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TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Functional operation	3.4	4.6.2

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One mount shall be selected at random as a special sampling from each 100 produced or from each month's production whichever occurs first. The sample shall meet the requirements and tests in Table II.

TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.2	Applicable drawings - visual
302. General specification	3.3	MIL-F-13926 - visual

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when

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non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.1 Inspection equipment design requirements.- Special test equipment to test the functional operation of the mount, including operating instructions, are listed on the Final Test Fixture Drawing, D8614915.

4.6 Test Methods and Procedures.

4.6.1 Operating temperatures.- The temperature cycling for the functional tests of the mount at the specified temperature extreme shall be conducted in accordance with 3.4.1 and the "Test Facilities" of Specification MIL-F-13926. When the mounts are thermally stabilized at minus 40 degrees and plus 150 degrees Fahrenheit, respectively they shall be subjected to the test specified in 4.6.2.

4.6.2 Functional operations.- This test shall be performed by using the special testing equipment specified in 4.5.2.1. The gage shall be inserted into the mount and locked in place by turning the barrel handle of the mount. Hand force shall be used to attempt to shake the gage within the mount. It shall remain rigid with no apparent movement. The fit between the mount collar and gage mating surfaces shall be flush. The handle should not release the device until the spring loaded button is depressed and the handle moved to the unlocked position. The spring catch on the mount shall prevent the periscope from falling free of the mount.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet P8270627. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- Mounts, M104, M104A1 and M104A2 are intended to mount periscopes of the M28 Series.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable stock number.

Custodian:

Army - MU

Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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